2024 4th Asian Conference on Innovation in Technology (ASIANCON)
Pune, India. Aug 23 - 25, 2024

Development of Women's Safety System Using Iot and Taser Technology

Kruthika N M
Department of Electronics and Communication
Rajalakshmi Engineering College
Chennai, Tamil Nadu, India
nmkruthika@gmail.com

Kavipriya S
Department of Electronics and Communication
Rajalakshmi Engineering College
Chennai, Tamil Nadu, India
kavipriya282002@gmail.com

Harshini P B
Department of Electronics and Communication
Rajalakshmi Engineering College
Chennai, Tamil Nadu, India
harshuu1402@gmail.com

H. Heartlin Maria
Department of Electronics and Communication
Rajalakshmi Engineering College
Chennai, Tamil Nadu, India

Priyanka Devi
Department of Electronics and Communication
Rajalakshmi Engineering College
Chennai, Tamil Nadu, India

2024 4th Asian Conference on Innovation in Technology (ASIANCON) | 979-8-3503-5421-8/24/$31.00 ©2024 IEEE | DOI: 10.1109/ASIANCON62057.2024.10837901

***Abstract* — The women safety device is an innovative IoT-based safety system designed to monitor key physiological parameters, such as heartbeat and EEG, and coupled with a panic switch, GSM, GPS, and IoT modules. This system provides real-time monitoring of vital signs, enabling swift detection of distress or emergencies. In the event of abnormal conditions or manual activation through the panic switch, the system triggers immediate alerts via the GSM module to predefined contacts, accompanied by real-time GPS location tracking. The IoT module facilitates seamless connectivity, enabling remote monitoring through a centralized platform. The women safety device prioritizes emergency response, user privacy, data security, and compliance with regulatory standards. It aims to empower individuals and contribute to a safer and more secure environment for women. A GPS-built women's sanctuary system with dual security elements is proposed to help overcome this issue. A woman can press the prepared switch when she is in danger and needs to act to protect her. When a woman, child, or senior is harassed or in distress, the gadget notifies the family and appropriate authorities of the woman's current position. An electric shock generator incorporated into the panic button on the gadget is a peculiar feature. In conclusion, the women safety device is a comprehensive solution to ensure the safety and well-being of women in today society.**

Keywords—Women, Panic switch, GSM (Global System for Mobile Communication), GPS (Global Positioning System), EEG (Electroencephalogram), Security, Monitoring

I. INTRODUCTION

Women's security and safety are given top priority in the Women's safety system. In the event of abnormal situations, it can be manually activated via the panic switch or it can employ real-time monitoring capabilities to detect distress signals or emergencies. In either case, quick notifications are sent through the GSM module to specified contacts. For prompt assistance, the technology additionally offers accurate real-time GPS location tracking. Safety device is a symbol of empowerment rather than merely a reaction tool. Through a centralised platform, remote monitoring becomes not only possible but also a reality thanks to the IoT module's smooth connectivity. It represents the shared desire for a more secure and safe environment for women and goes beyond the boundaries of technology. Safety device was developed because of a thorough study of the difficulties and threats to women's safety that they encounter. Conventional approaches to safety, like self-defence classes or depending on the police, frequently fail to offer women in need of help prompt and complete protection. By providing a comprehensive safety solution that enables women to take charge of their safety and get prompt assistance when needed, Safety device aims to overcome these constraints. To guarantee smooth connectivity and communication, the system makes use of the capabilities of GSM, GPS, and IoT modules. When an emergency arises, the system uses the GSM module to immediately notify pre-designated contacts. Real-time GPS location tracking is also included to enable prompt aid and response. Furthermore, the IoT module enables remote monitoring and management through a centralized platform, enhancing the effectiveness and efficiency of the safety system.

II. RELATED WORKS

Monisha, D. G., Monisha, M., Pavithra, G., & Subhashini, R. (2016) introduced a security device called FEMME, designed for women, that can be easily installed on smart phones for emergency assistance [1]. The device functions by squeezing the volume key and control button, showing four basic symbols: sound recorder, SOS message, video recorder, and covered camera locator. The device also has two buttons for emergency assistance, one for the crisis button and the other for covered camera detection. Paradkar, A., & Sharma, D. (2015) proposed a demonstration including auto-accepting call modules; spy camera detection modules, fake call instrument modules, and generate electric stun modules to improve security for women [2]. Harikiran, G. C., Menasinkai, K., & Shirol, S. (2016) proposed a device to provide protection, alert victims, and send data to nearby individuals [3].

The Savvy Band uses Bluetooth Mobility (BLE) sensors, movement sensors, and temperature sensors, making it reliable and efficient. It also communicates with family, police, and others. Sanket Chikhale, et.al (2022) "IOT BASED EMERGENCY BUTTON FOR WOMEN SAFETY" [4].This study introduces IoT-based smart home security systems, like Nest Lab's smoke and carbon monoxide alarm, are a growing trend in the security industry. They are constantly connected and communicate using wired

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and wireless networks, providing real-time alerts of potential danger.

The study by Sharad Saxena et al. (2023) presents a wearable safety device called the Women Safety Gadget (WSG) based on IoT and Cloud Technology [5]. The device detects unsafe conditions by detecting a finger pressure on an artificial nail, diffuses a pepper spray, and notifies the nearest contacts and police station. The device has a response time of 1000 ms and an activation time of 2.079 s, making it a potential solution for women's safety. The study by Rajamani et al. (2022) aims to enhance the security of women in the IT industry by developing an enhanced wearable strap using IoT technology [6]. The device, which includes sensors, computing power, and software, uses vibration to keep individuals aware and alert. The device uses GPRS and GSM Neo 6 m to determine location, and a pulse sensor detects the person's pulse rate. If the vibration does not stop within specified time, an emergency alert is sent to predefined contacts on the Arduino board.

III. PROPOSED METHODOLOGY

A. *Problem Definition*

Threats to women's physical and mental health include harassment, assault, and kidnapping are becoming more prevalent. The frequency of these crimes is still on the rise, despite developments, thus creative safety measures are required. Women's vulnerability in public and private areas is further exacerbated by societal norms and cultural considerations. The development of Internet of Things (IoT)-based safety devices and systems is imperative due to the major gap in the provision of suitable protection and support mechanisms for women. These gadgets provide real-time communication, monitoring, and intervention features to reduce dangers and guarantee women's and children's safety and security in a variety of settings. The complexity and urgency of women's safety challenges, as well as the need for technical interventions, are highlighted in this problem statement.

The Existing system presents a safety gadget for women using a NodeMCU microcontroller for research. The device uses Internet of Things communication to send alerts to secure channels, with a distress algorithm. An IoT platform called Thing Speak allows for continuous monitoring of health data every 30 seconds. However, drawbacks include a lack of defined threshold values for the distress algorithm, lack of detailed false alert management, unclear data usage and privacy, and overlooking user interface and accessibility aspects, especially in emergency situations.

Our Safety system is a contemporary remedy created to safeguard the security and well-being of women is the women safety gadget. It recognises emergencies and provides real-time GPS location tracking, beginning when sensor readings above threshold values. It does this by utilising sophisticated IoT technologies, such as panic buttons, heartbeat sensors, EEG monitors, GSM, GPS and IoT modules. An Additional switch on the gadget can be used to manually sound an alert in an emergency. The system places a high priority on user privacy, emergency response, and regulatory compliance, empowering women to take charge of their own safety. The gadget provides customers with proactive safety measures, prompt GSM notifications, IoT integration for remote monitoring, and immediate alarms.

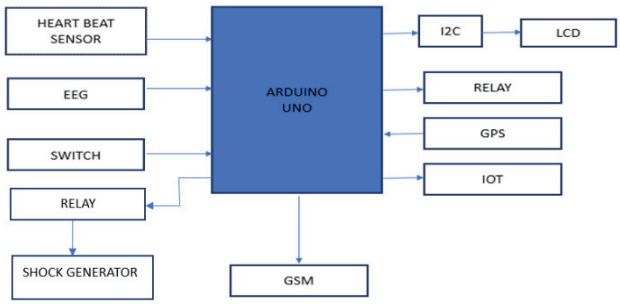


Fig. 1. Block diagram of women safety device

B. *System Architecture*

The Arduino UNO was selected for its ability to offer wireless connectivity, facilitating instant communication between the device and external systems. It comes with built-in Wi-Fi feature that permits data transmission over the internet or local network. Moreover, the Arduino' low power consumption and cost-effectiveness make it a perfect option for portable and battery-operated devices. The integration of Arduino includes linking it with other hardware components like the GPS module and GSM module. The microcontroller communicates with these modules using appropriate protocols such as Serial Peripheral Interface (SPI) or Universal Asynchronous Receiver-Transmitter (UART) to ensure smooth data exchange and synchronization.

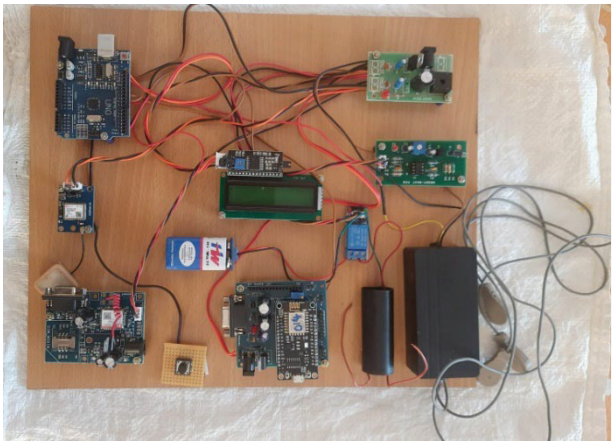


Fig. 2. Prototype of women safety system

The placement of the heartbeat rate sensor and EEG within the device is strategic for monitoring the user's heartbeat and detecting sudden changes. It can recognize potential dangers like physical attacks or falls. Upon detecting a threatening movement or impact, the device activates the emergency alert system, triggering the GPS module to transmit the location coordinates. Simultaneously, the taser system is also activated for immediate self-defence. The GPS module plays a vital role in obtaining precise real-time location information. It continuously tracks the user's position using satellite signals, enabling emergency responders or contacts to accurately locate the user in distress situations. The GPS data is processed, combined with other relevant details, and then transmitted via the GSM module.

This device has a panic button that allows the user to manually trigger the safety mechanism whenever they feel in danger. According to Fig. 3, as soon as someone presses the panic button, the taser, GPS, and GSM modules are all

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initialised, and the locations are gathered and forwarded to the emergency contacts. Our safety system also includes an electroencephalogram (EEG) sensor and a heartbeat sensor to increase safety and provide non-manual device initiation.

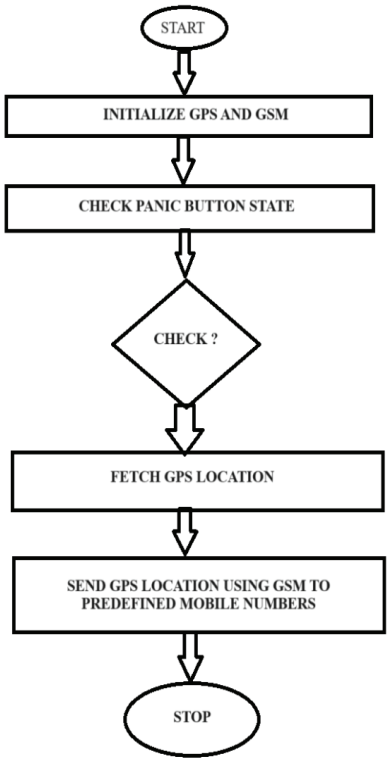


Fig. 3. Flowchart of panic button mechanism

To turn on a heartbeat sensor, power the board from a 3-5.5V source and pull the Enable (EN) pin high. Next, lightly press the tip of your forefinger against the sensor's face. Avoid applying excessive pressure to the sensor with your finger and keep it still. The LED will begin to blink in time with your heartbeat after a few seconds of the circuit stabilising. To measure the heart rate in beats per minute (BPM), you can feed the output signal to the microcontroller's digital I/O or ADC input pin. This sensor determines whether a person's heart rate has risen or fallen below a predetermined threshold value for about 60 seconds as specified in the Arduino UNO. The position of the user is then gathered and relayed to their designated contacts using the GPS and GSM modules. The shock generator is turned on at the same time as this process

The sensor detects a person's heartbeat and compares it to the threshold values, as seen in Fig. 4. It signals an emergency with the message "low heart beat" if the pulse rate is between 10 and 20 beats per minute. An emergency alert indicating a high heart rate is sent if the detected pulse rate is greater than or equal to 80 beats per minute.

Electroencephalogram (EEG) sensors require conductive gel to make low-impedance electrical contact with the skin. We provide a gel-free, non-contact EEG sensor that connects to the skin capacitively using an on-board electrode. The device consists of a headset, an ear clip, and a sensor arm. The reference and ground electrodes of the headset are on the ear clip, while the EEG electrode is on the sensor arm, positioned on the forehead above the eye (FP1 position). Using a single AAA battery, it can run for eight hours.

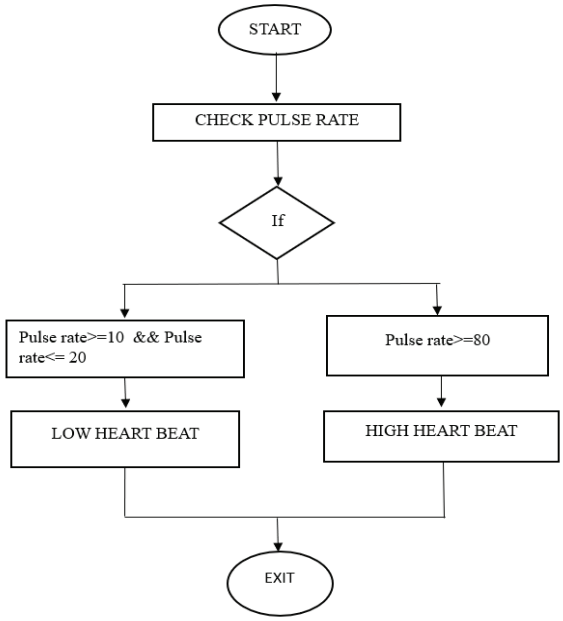


Fig. 4. Working of heart beat sensor

The EEG sensor in Fig. 5. detects electrical waves in the brain and compares them to the threshold value. An emergency alert indicating an abnormal EEG rate is sent to the person's emergency contact and an electric shock generator is also initiated if the EEG sensed value is more than 3 volts.

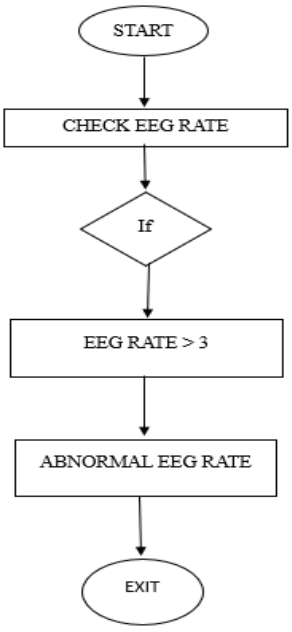


Fig. 5. Working of EEG sensor

Working procedure: This device enables the user to activate it both manually and automatically.

Step 1: Get going.

Step 2: Turn on the power supply

Step 3: Press the emergency button.

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Step 4: If GPS picks up a signal, it will begin determining the victim's current latitude and longitude and use the GSM module to send an SMS to the registered mobile number.

Step 5: The device obtains the most recent location from GPS and transmits it to the GSM module if either a high or low heart rate is detected.

Step 6: The device obtains the last location from GPS and transmits it to the GSM module if an abnormal EEG rate is detected.

Step 7: The victim's last location is tracked by an IoT module, and the webpage is updated with that information.

Step 8: Turn on the Taser system to stun the assailant.

Step 9: Come to an end.

IV. RESULTS AND DISCUSSION

The primary aim of this project is to provide a comprehensive solution for ensuring the safety and well-being of women today, leveraging IoT technology to monitor key physiological parameters and facilitate swift response in emergency situations. The project has been tested and the results have been attached below.

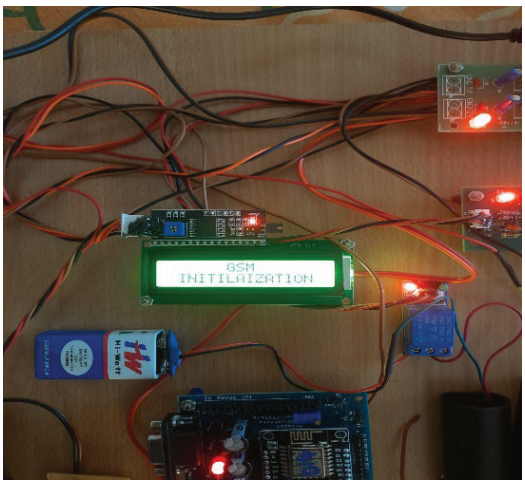


Fig. 6. Initialization of GSM

Once the device has been set up, GSM will be initialized and the display of the system will be as shown as in Fig. 6. and Fig. 7.

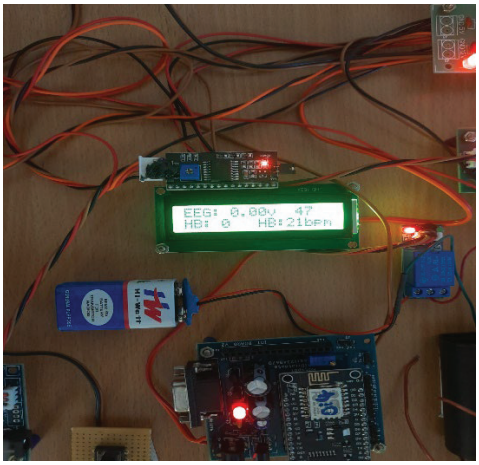


Fig. 7. LCD display of the system

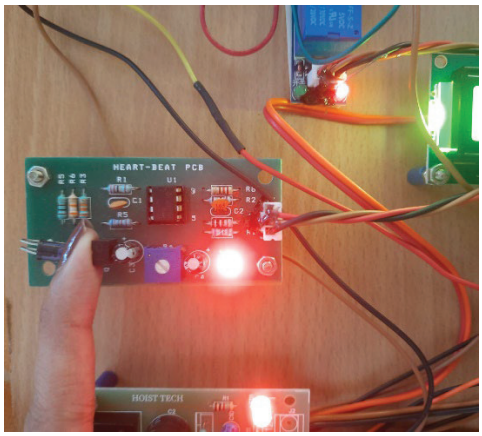


Fig. 8. Sensing of pulse rate using heart rate sensor



Fig. 9. Low heart beat alert message



Fig. 10. High heart beat alert message

Whenever a low heartbeat or high heartbeat is detected (whose threshold has been defined in the module) it records the reading and sends the message to the emergency contacts that has been saved earlier in our system as shown in Fig. 9. and Fig. 10. Along with this, the user's location consisting the longitude and latitude is sent to the emergency contacts immediately using the GPS and GSM module as show in Fig. 11.

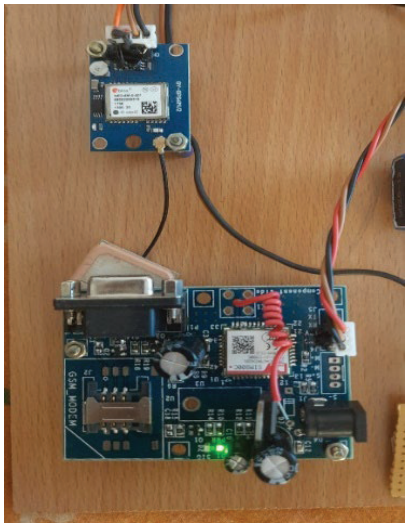


Fig. 11. GPS and GSM module

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Similarly, whenever it detects an abnormal EEG (whose threshold has been defined in the module) or when the panic button is pushed, an alert message consisting of the user's location is send to the predefined emergency contacts with the help of GPS and GSM module as shown in the Fig. 12. and Fig. 13.

TABLE I.

Preference	Sensor	Location Detection	SMS	Taser activation
Proposed System with EEG and Heart Beat sensor	Yes	Yes	Yes	Yes
[1] Using Mobile Application	No	Yes	Yes	No
[2] Using Finger Pressure	Yes	Yes	Yes	No



Fig. 12. Abnormal EEG alert message



Fig. 13. Panic button triggered alert message

This system also has an IOT module shown in Fig. 14. where the user's network can be connected and the entry of all the alert messages sent along with its sent time and location is stored in cloud as shown in Fig. 15.

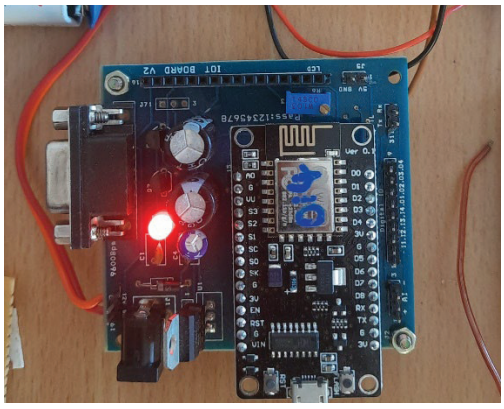


Fig. 14. IOT module

V. CONCLUSION

Women safety device is an innovative approach to women's protection that uses Internet of Things (IoT) technology to solve the critical issue of protecting women in contemporary society. This device provides a layered safety system that goes beyond simple monitoring by carefully combining a variety of state-of-the-art components, including ESP8266, Arduino Uno, heart rate sensor, EEG sensor, GSM, GPS, and Taser System.

- It guarantees prompt reaction in emergency scenarios by monitoring vital signs in real-time.
- Sends out instant notifications within seconds in response to abnormal conditions or manual intervention via the panic button.
- Users are empowered with proactive safety measures and their trust is increased by the system's centralised platform and seamless connection, which allow for 24/7 remote monitoring.
- It sends health related data which is almost 95% accurate.

Furthermore, it places a high priority on user privacy, data security, and regulatory compliance, demonstrating its dedication to protecting people in all respects. It exemplifies development, empathy, and empowerment by representing the shared desire for a more secure and safe environment for women and is essentially a symbol of empowerment that goes beyond the boundaries of technology, providing not just a safety answer but also a hope for a safer and better future for women everywhere.

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